Project Documentation Report

Automating Data Import and Relationship Mapping using Import Sets and Dot Walking

Date	05 April 2026
Team ID	SWTID-2026-5878
Project Name	Automating Data Import and Relationship Mapping using Import Sets and Dot Walking
Maximum Marks	2 Marks

Team Members

S.No	Name	Reg No	Role
1	Sairam M	812823205060	Team Leader
2	Karpaga Vinayagam S	812823205032	System Admin
3	Manuneethi Cholan S	812823205047	System Admin
4	Saravanan R	812823205064	QA Tester
5	Rada Krishnan S	812823205055	System Admin

1. INTRODUCTION

1.1 Project Purpose

The primary purpose of this project is to eliminate the manual overhead, data latency, and human error associated with onboarding new employee records into the IT service management platform. Furthermore, it aims to significantly reduce the Average Handle Time (AHT) of IT support tickets. Currently, IT Support Agents must manually navigate away from an Incident to look up an employee's department and managerial escalation path. By automating the data pipeline and surfacing this contextual data directly on the Incident form, the project intends to create a seamless, highly efficient workflow for both System Administrators and IT Support teams.

1.2 Project Overview

This project delivers an automated, end-to-end data ingestion and UI enhancement solution within ServiceNow, leveraging out-of-the-box, low-code platform capabilities.

The initiative is divided into two core phases:

- **Phase 1: Automated Data Pipeline (Backend):** We will establish a robust data ingestion engine using **System Import Sets**. External employee data (Name, Email, Department, Manager) provided via CSV/XLSX will be staged in a custom `import_employee` table. A **Transform Map** will then be configured to intelligently push this data into the production `sys_user` table. This map will use Auto Map for standard fields and Mapping Assist to accurately resolve complex reference fields (linking the employee to the correct Department and Manager `sys_id` records).

- **Phase 2: Dynamic UI Enhancement (Frontend):** Once the data architecture is solid, we will enhance the Incident management experience. Using **Dot-Walking**, the Incident form will be configured to automatically pull and display the assigned user's Department and Manager's Email in real-time. This ensures that the moment a ticket is assigned, the support agent has immediate visibility into the user's organizational context without writing any custom client-side scripts.

1.3 Key Objectives & Expected Outcomes

- **Data Integrity & Automation:** Achieve a 100% automated sync of external HR data into ServiceNow, ensuring accurate reporting and user management.
- **Zero-Code Architecture:** Utilize platform-native features (Import Sets, Transform Maps, Dot-Walking) to ensure the solution is highly scalable, maintainable, and upgrade-safe.
- **Enhanced Agent Experience:** Empower IT Support Agents with immediate, zero-click visibility into escalation paths (Manager) and business context (Department) directly on the active Incident form.
- **Strict Security Compliance:** Ensure that all data loads and UI modifications strictly adhere to existing Role-Based Access Controls (RBAC) and Access Control Lists (ACLs).

2. IDEATION PHASE

2.1 Problem Statement Definition

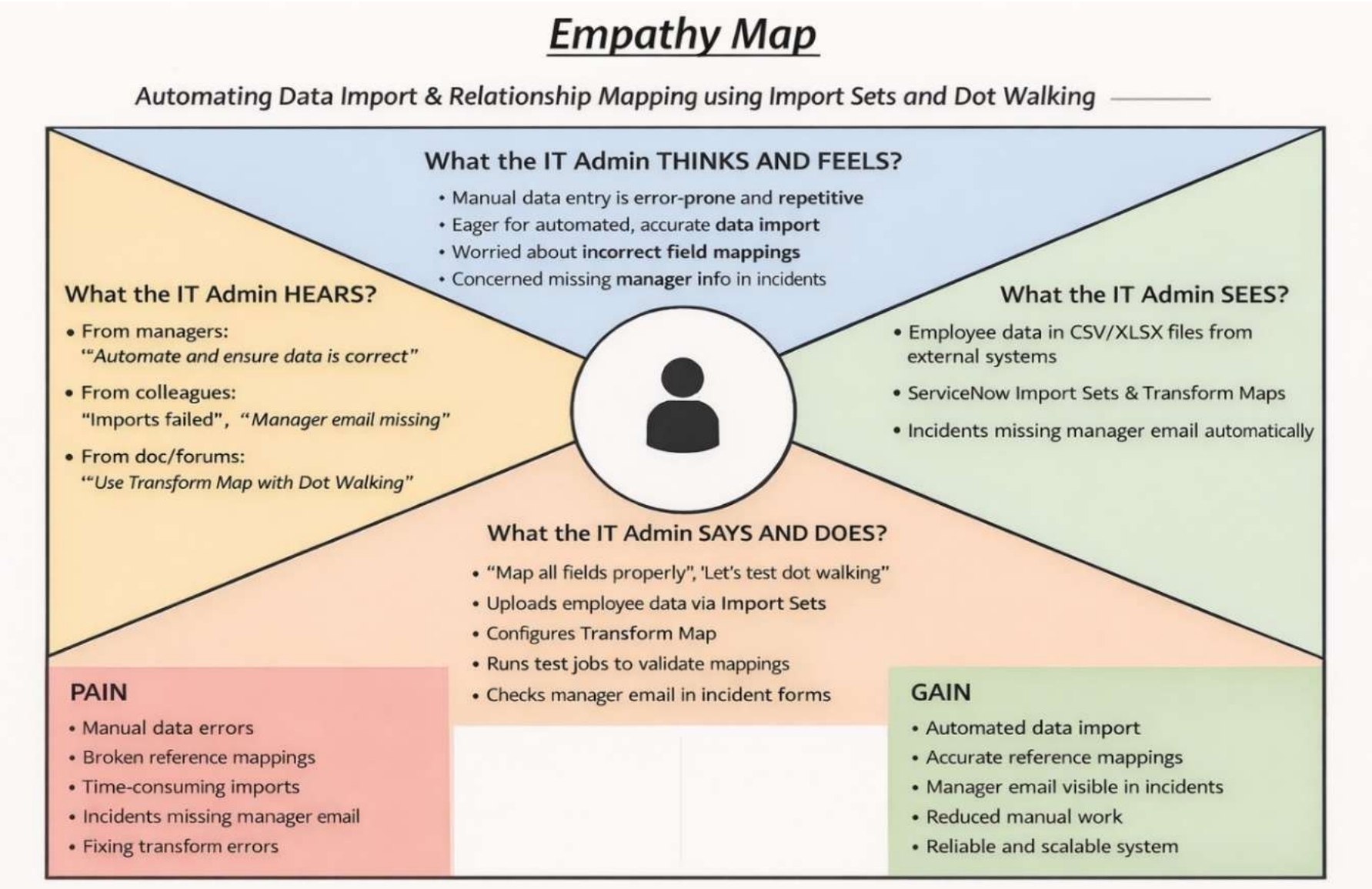
Our Problem Statement:
Our organization wants to regularly import a list of new employees with their details, such as Name, Email, Department, and Manager, from an external system. Once imported, when an Incident is assigned to an employee, the Incident form should automatically show the manager's email of the employee's department.

I am	IT Administrators, ServiceNow Developers, HR Data Managers	IT Administrators, ServiceNow Developers, HR Data Managers, and Incident Managers
I'm trying to	regularly import a list of new employees (Name, Email, Department, Manager)	regularly import a list of new employees (Name, Email, Department, Manager) and ensure that, when an Incident is assigned to an employee, the Incident form automatically shows the manager's email of the employee's department.
but	Manual data import is time-consuming, prone to errors,	manual data import is time-consuming, prone to errors, and lacks automated relationship mapping for accurate department and manager references, leading to incomplete or incorrect data.
because	employee data is not automatically synchronized between external	employee data is not automatically synchronized between external systems and ServiceNow, requiring frequent manual processing and missing reference mapping during incidents.
which makes me feel	frustrated, overwhelmed	frustrated, overwhelmed by repetitive tasks, concerned about incorrect data guiding incident management, and eager for a streamlined, automated solution.
	frustrated, overwhelmed	**Overwhelmed** by repetitive tasks, concerned about incorrect data guiding incident management, and eager for a streamlined, automated solution.

Problem Statements Identified:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A ServiceNow System Administrator	regularly sync new employee details (Name, Dept, Manager) from an external system into the platform...	the current method is manual, time-consuming, and prone to data relationship errors...	we lack an automated pipeline using Import Sets and Transform Maps.	overwhelmed by manual data entry and inefficient.
PS-2	An IT Support Agent	quickly identify an employee's department and manager to escalate an assigned Incident...	I have to manually navigate away from the Incident form to dig through the user's profile...	the Incident form does not automatically surface related reference data (dot-walking).	frustrated by unnecessary clicks and slowed down in my resolution time.

2.2 Empathy Map



2. Brainstorming and Idea prioritization

1. Problem Statement Automate employee data imports and display the manager's email on Incidents without manual entry.

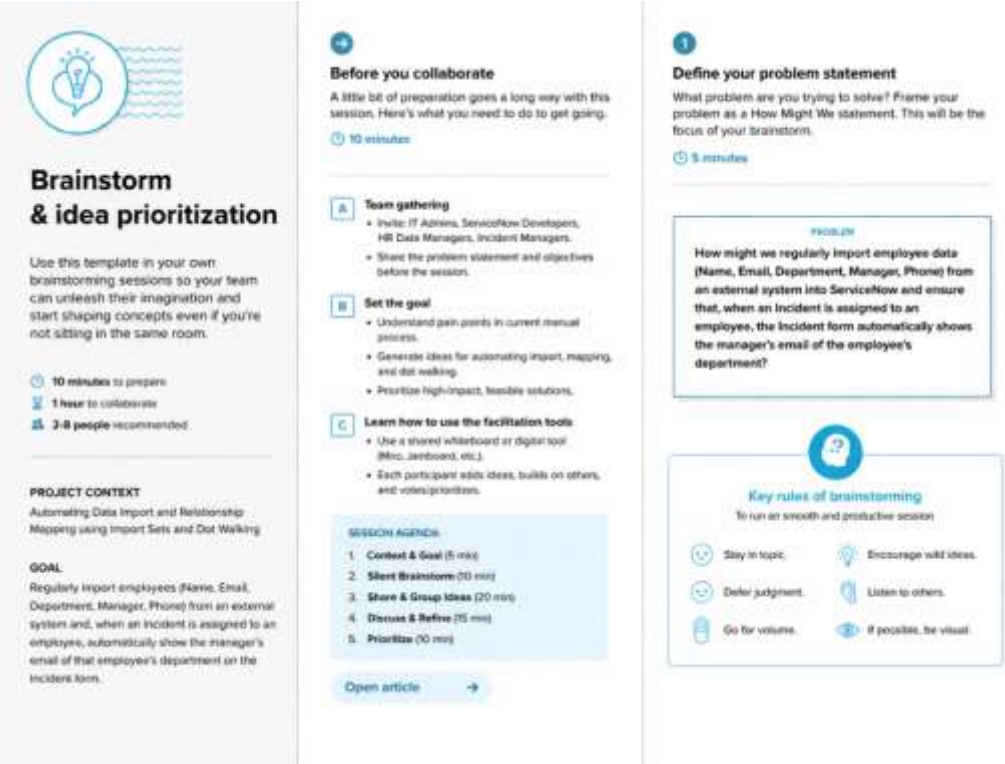
2. Brainstorming

- **Import:** Scheduled Import Sets vs. REST API
- **Display:** Form Layout Dot-Walking vs. Client Scripts?

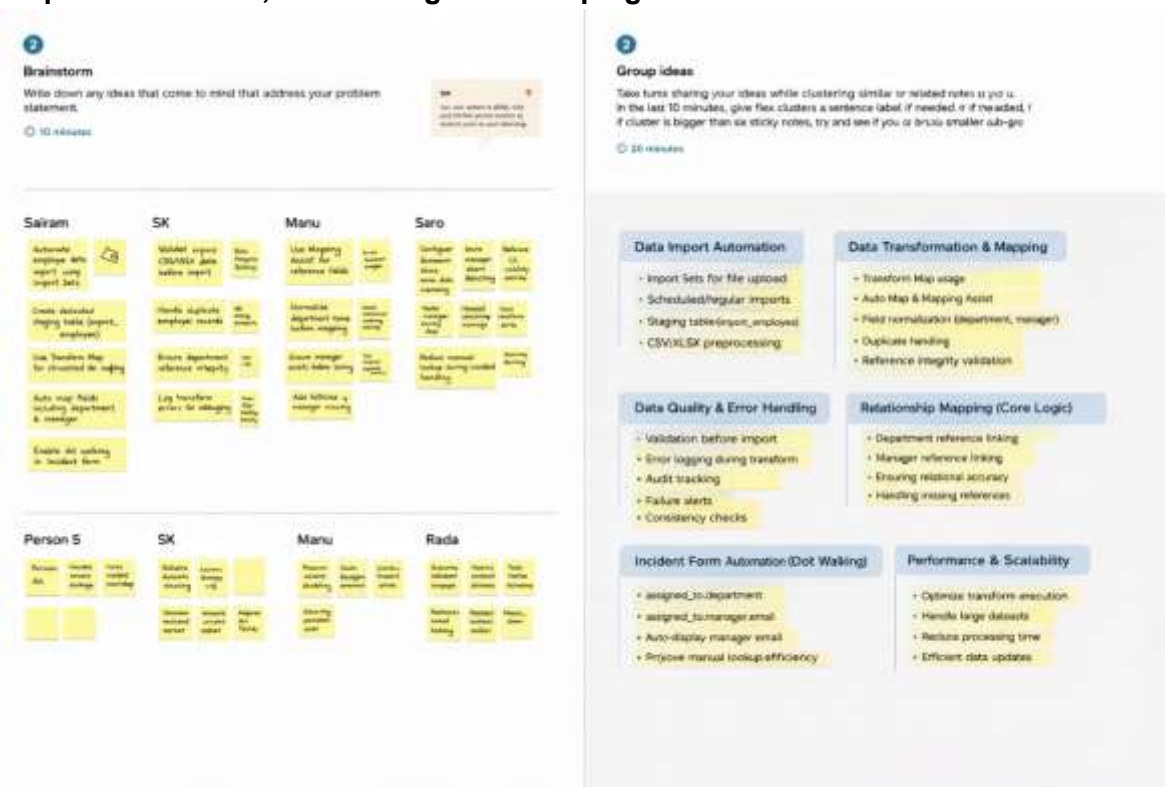
3. Prioritization

Selected Approach: Import Sets + Dot-Walking (Highest score for performance and ease of use).

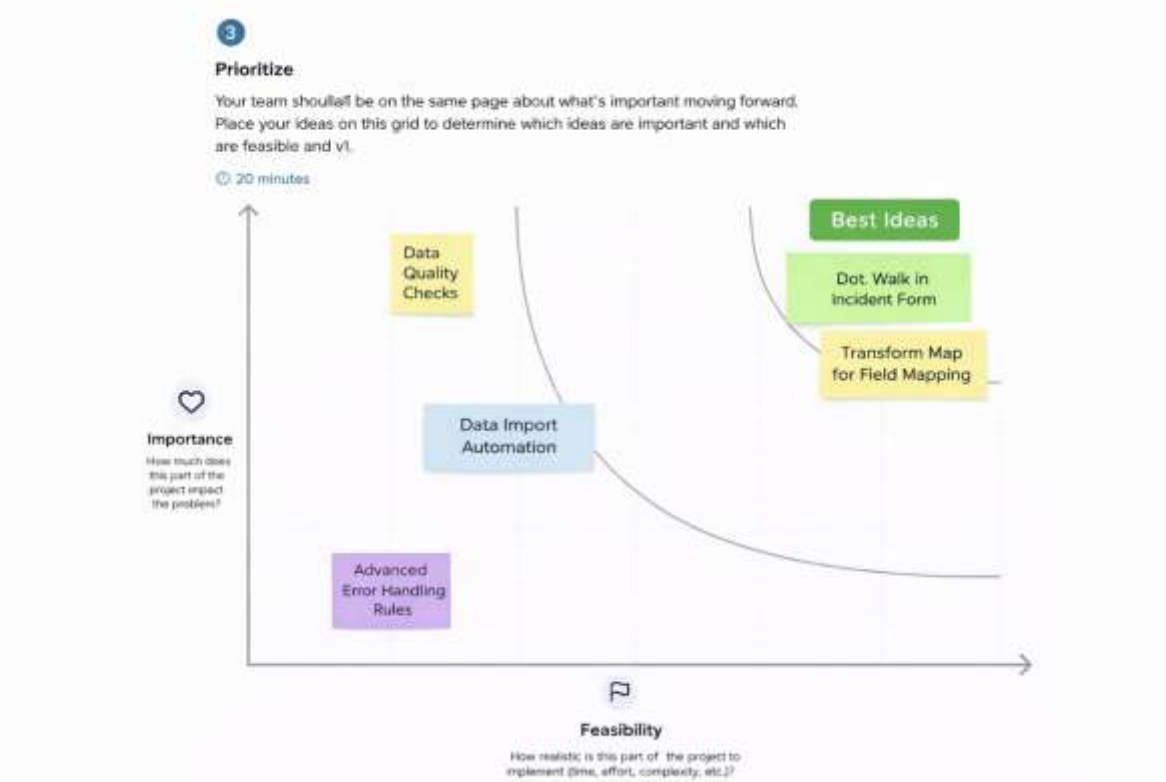
Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping

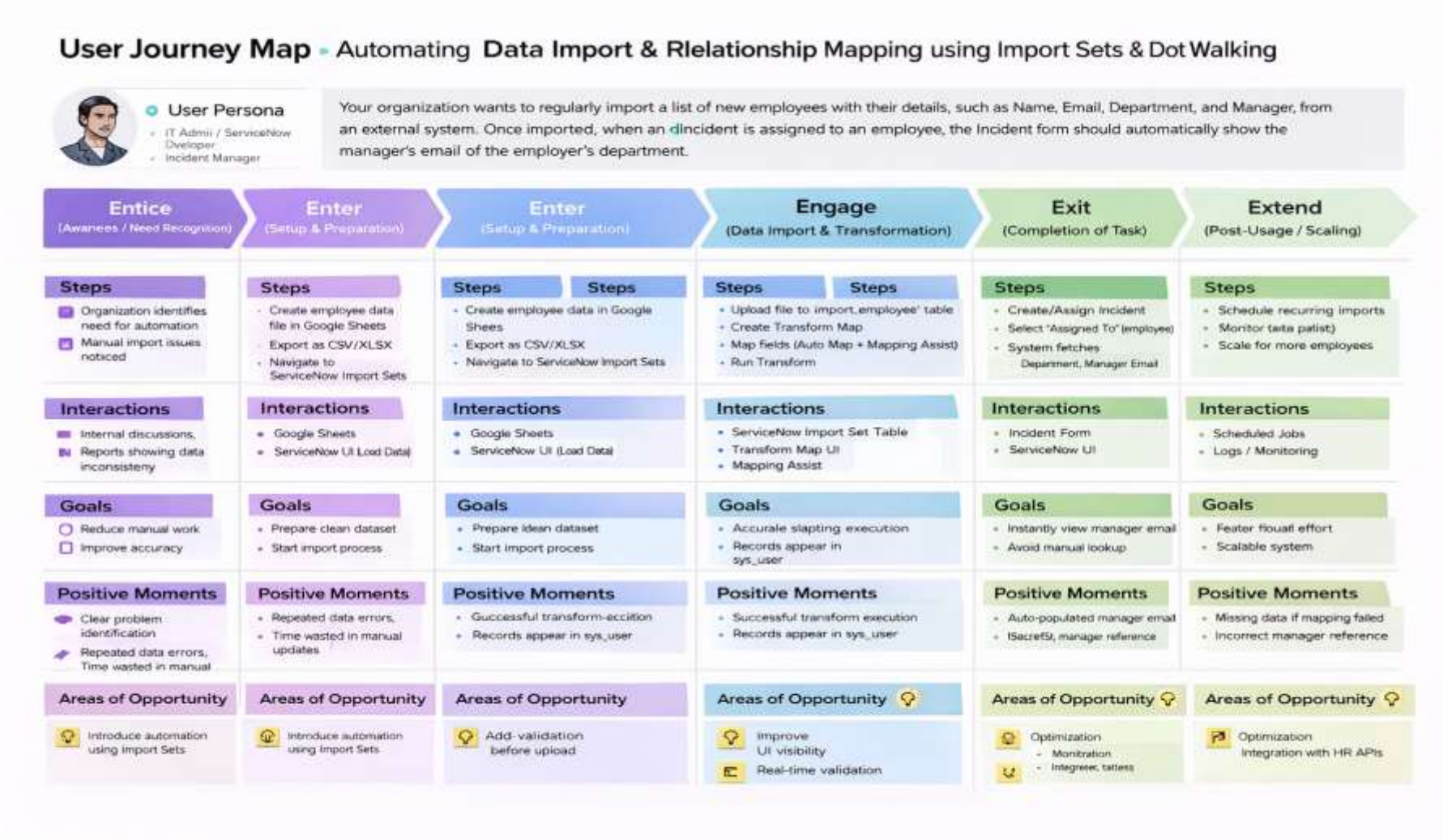


Step-3: Idea Prioritization



3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map



3.2 Solution Requirements

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**Employee Data Staging**	• Upload employee data (CSV/XLSX) via System Import Sets. • Automatically generate and populate the `import_employee` staging table.
FR-2	**Data Transformation & Mapping**	• Create a Transform Map linking import_employee to the sys_user table. • Map standard string fields (Name, Email, Phone) using Auto Map. • Configure Mapping Assist for reference fields (Department, Manager) to find matching sys_ids.
FR-3	**Transform Execution & Validation**	• Execute the Transform Map engine. • Validate successful record insertion and relationship linking in the sys_user table.
FR-4	**Incident Form Automation**	• Configure the Incident form layout. • Add dot-walked fields (Assigned to.department, Assigned to.email, Assigned to.manager) to the form view.

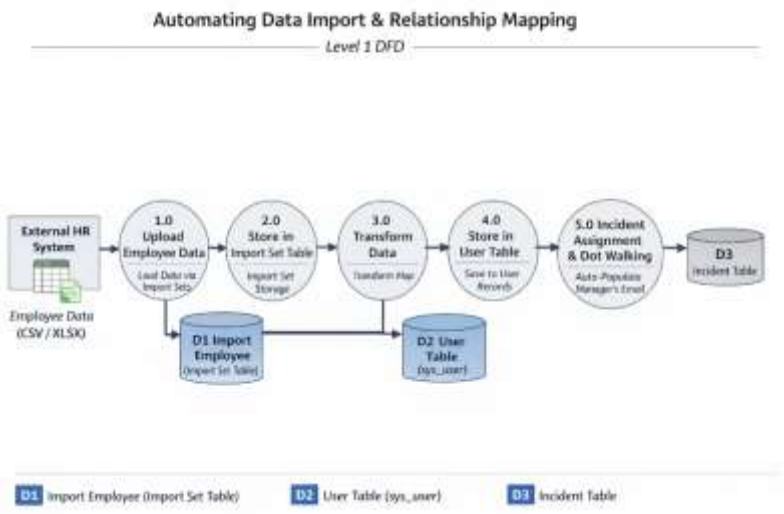
Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

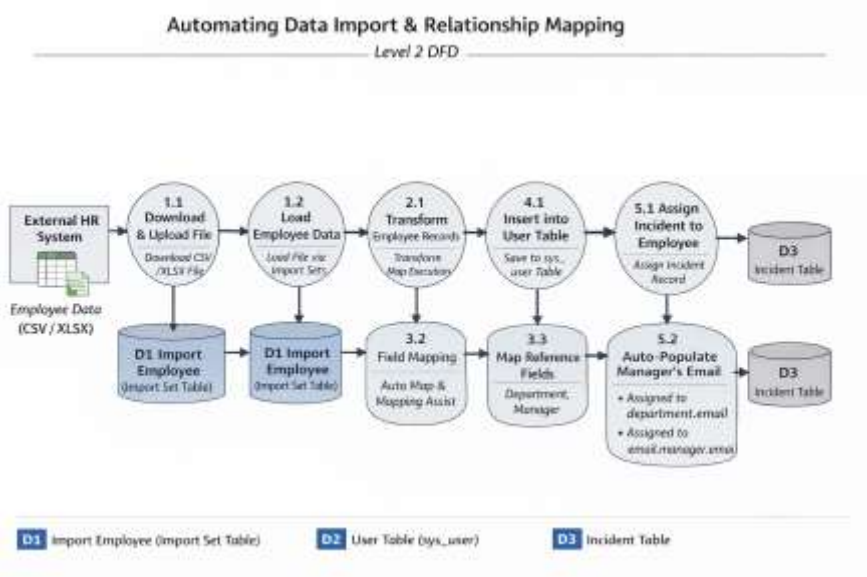
NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Incident form must auto-populate the dot-walked fields seamlessly and immediately when the "Assigned to" user is selected, without requiring a page reload.
NFR-2	**Security**	Only users with `admin` or designated import roles should have access to load data and configure Transform Maps. Data displayed on the Incident form must adhere to existing ServiceNow Access Control Lists (ACLs).
NFR-3	**Reliability**	The data import process should handle discrepancies gracefully. For example, if a provided manager does not exist in the system, the transform should log the error but continue processing the rest of the batch.
NFR-4	**Performance**	Processing the import sets should not cause noticeable degradation to platform performance. Dot-walked client-side lookups on the Incident form must load instantaneously.
NFR-5	**Availability**	The data import module and the modified Incident form must maintain the standard high availability of the core ServiceNow platform.
NFR-6	**Scalability**	The Transform Map logic must be capable of efficiently processing both small daily updates and large bulk imports (e.g., thousands of rows during an acquisition or major onboarding event).

3.3 Data Flow Diagrams

Level 1 DFD



Level 2 DFD

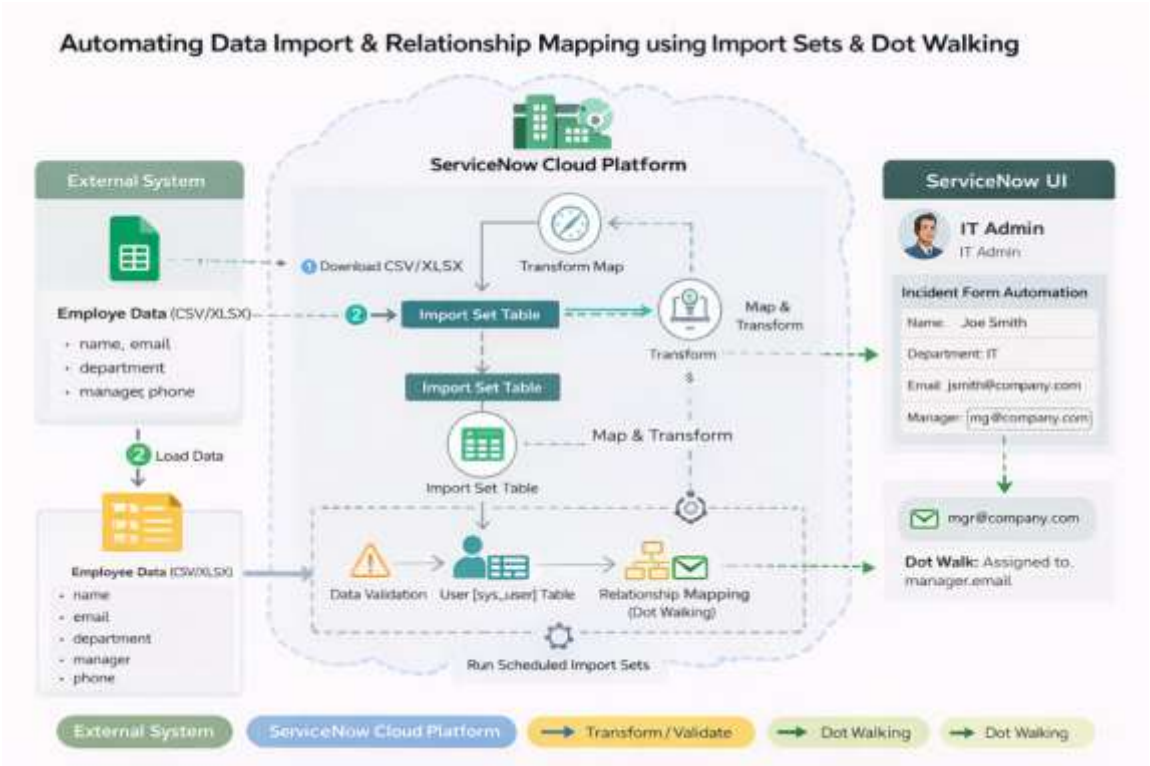


User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
ServiceNow Admin	Data Import Setup	USN-1	As a ServiceNow Administrator, I want to load a CSV file of employee data into an Import Set table named import_employee so that the external data is staged for processing.	1. CSV is successfully uploaded via Load Data. 2. import_employee table is automatically generated with the raw data.	High	Sprint-1
ServiceNow Admin	Data Transformation	USN-2	As a ServiceNow Administrator, I want to create a Transform Map linking the import_employee source table to the sys_user target table so that I can define how the data is migrated.	1. Transform map is created and saved. 2. Source and Target tables are correctly defined.	High	Sprint-1
ServiceNow Admin	Data Transformation	USN-3	As a ServiceNow Administrator, I want to map all standard and reference fields (department, manager) using Auto Map and Mapping Assist so that records are accurately linked in the system.	1. Name, email, and phone map cleanly. 2. Department and Manager reference fields look up and link to the correct sys_ids in the target tables.	High	Sprint-1
ServiceNow Admin	Data Transformation	USN-4	As a ServiceNow Administrator, I want to run the Transform Map and verify the output so that I know the employee records are successfully inserted into the user table.	1. Transform runs without data truncation errors. 2. Employee profiles are visible and accurate in the sys_user table.	High	Sprint-1
ServiceNow Admin	Form Configuration	USN-5	As a ServiceNow Administrator, I want to configure the Incident form layout using dot-walking so that related user details are exposed on the form.	1. Assigned to.department, Assigned to.email, and Assigned to.manager are added to the Incident form view.	High	Sprint-1
IT Support Agent	Incident Management	USN-6	As an IT Support Agent, I want the assigned employee's department, email, and manager to auto-populate when I assign an incident, so I have immediate context without opening their profile.	1. Selecting an employee in the "Assigned to" field immediately displays their department, email, and manager's email on the form.	High	Sprint-1

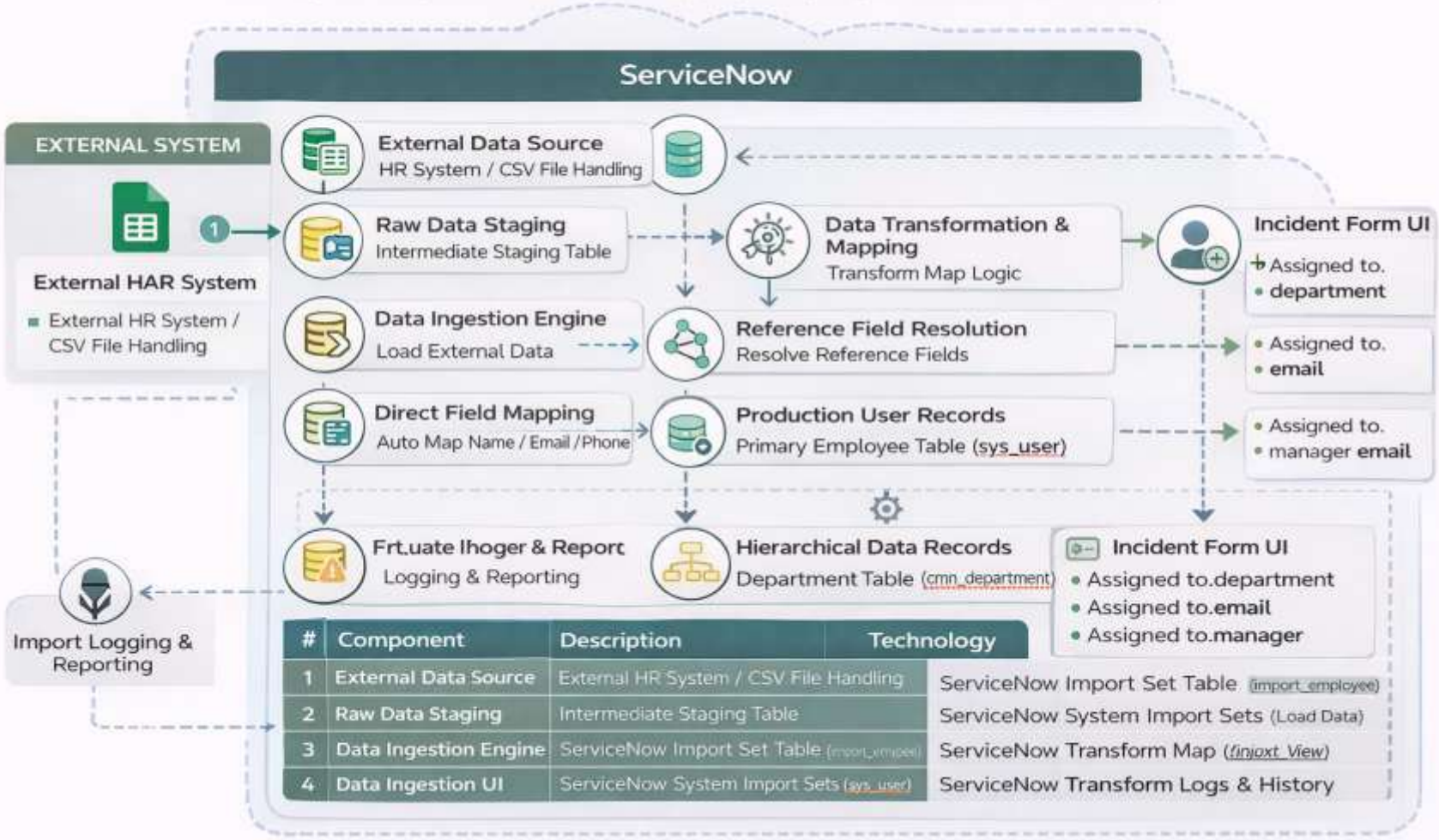
3.4 Technology Stack

Technical Architecture:
Following is the Technical Architecture DIagram of our project



Technical Architecture on ServiceNow

Automating Data **Import** & Relationship Mapping using **Import Sets & Dot Walking**



#	Component	Description	Technology
1	External Data Source	External HR System / CSV File Handling	ServiceNow Import Set Table (import_employee)
2	Raw Data Staging	Intermediate Staging Table	ServiceNow System Import Sets (Load Data)
3	Data Ingestion Engine	ServiceNow Import Set Table (import_employee)	ServiceNow Transform Map (*finjoxt_View*)
4	Data Ingestion UI	ServiceNow System Import Sets (sys_user)	ServiceNow Transform Logs & History

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Table-1: Components & Technologies:

S.No.	Component	Description	Technology
1.	**External Data Source**	Configuration to securely ingest the initial list of new employee details.	External HR System / CSV File Handling
2.	**Raw Data Staging**	An intermediate, transient storage table used to hold raw data from the CSV file.	ServiceNow Import Set Table (`import_employee`)
3.	**Data Ingestion Engine**	Platform mechanism used to load external data into the staging table.	ServiceNow System Import Sets (Load Data)
4.	**Data Transformation & Mapping**	Core logical component that defines how raw staging data is converted and mapped to target production records.	ServiceNow Transform Map
5.	**Direct Field Mapping**	Automated process to link simple text fields like Name, Email, and Phone.	ServiceNow Auto Map
6.	**Reference Field Resolution**	Logic to look up and link existing production records for complex relationships (Department, Manager).	ServiceNow Mapping Assist
7.	**Production User Records**	The platform's primary, persistent table for active employee profiles.	ServiceNow User Table (`sys_user`)
8.	**Hierarchical Data Records**	The production table defining the organization's department structure, required for valid relationship lookups.	ServiceNow Department Table (`cmn_department`)
9.	**Incident Form UI**	The user interface view of the Incident table, which is configured to display cross-referenced user details.	ServiceNow Form Design (Incident default view)
10.	**Dynamic UI Binding**	Platform-native, zero-code logic used to expose related data on the Incident form automatically.	ServiceNow Dot-Walking (`Assigned to.department`, `Assigned to.email`)
11.	**Import Logging & Reporting**	The system tool used to track the success or failure of each automated data load batch.	ServiceNow Transform Logs & History

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	**Open-Source Frameworks**	No external open-source frameworks are directly utilized for this configuration. The solution relies entirely on built-in, out-of-the-box platform capabilities.	ServiceNow Platform Native Features (Proprietary)
2.	**Security Implementations**	Access to upload data and configure Transform Maps is strictly limited via Role-Based Access Control (RBAC). Visibility of the dot-walked fields on the Incident form is governed by existing platform Access Control Lists (ACLs).	ServiceNow ACLs, RBAC (`admin`, `import_admin` roles)
3.	**Scalable Architecture**	The solution leverages ServiceNow's multi-instance cloud architecture. The Transform Engine uses an asynchronous staging table model, allowing it to scale effortlessly from processing 10 records to 10,000 records without blocking platform operations.	ServiceNow Multi-Instance Architecture, Async Import Sets
4.	**Availability**	The data pipeline and modified Incident forms inherit the enterprise-grade High Availability (HA) of the core ServiceNow platform, ensuring the feature is always online.	ServiceNow Advanced High Availability (AHA) Cloud Infrastructure
5.	**Performance**	Import processing uses staging (`import_employee`) to prevent database locking on the live `sys_user` table. On the UI side, dot-walking is processed server-side, entirely avoiding the need for slow, performance-heavy client-side scripting (like GlideAjax).	Transform Map Engine, Server-side UI Rendering (Dot-walking)

4. PROJECT DESIGN

4.1 Problem - Solution Fit

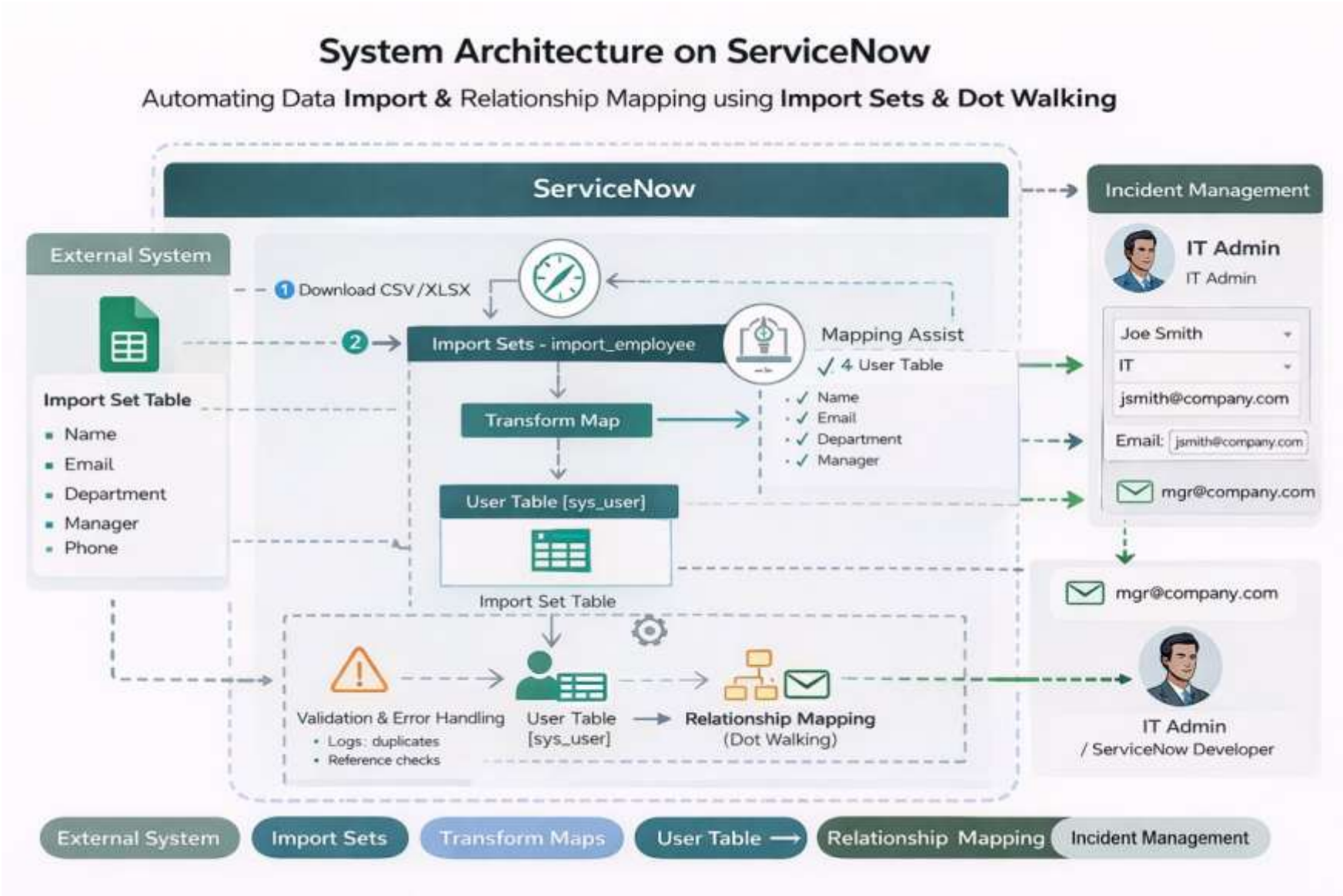
The following image showcases the fitness of our solution to the current problem.

4.2 Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	The organization currently lacks an automated way to sync new employee data from external HR systems into ServiceNow. Furthermore, IT Support Agents lack immediate visibility into an employee's department and manager when assigning incidents, leading to manual lookups and delayed resolution times.

S.No.	Parameter	Description
2.	Idea / Solution description	Develop an automated data pipeline using ServiceNow Import Sets and Transform Maps to regularly ingest and map external employee CSV data into the sys_user table. Enhance the user interface by configuring Dot-Walking on the Incident form so that selecting an assignee automatically surfaces their related department and manager details without requiring custom scripting.
3.	Novelty / Uniqueness	The solution leverages platform-native, low-code/no-code features (Transform Maps and Dot-Walking) rather than relying on complex custom Client Scripts or GlideAjax calls. This ensures a highly maintainable, upgrade-safe architecture that instantly binds hierarchical data to the user interface.
4.	Social Impact / Customer Satisfaction	High Internal Customer Satisfaction: IT Support Agents experience a significant reduction in friction and "clicks" required to find escalation paths. This immediate context accelerates incident routing and resolution, which in turn improves the overall IT support experience for all employees within the organization.
5.	Business Model (Revenue Model)	Cost-Saving / Internal ROI Model: While not a revenue-generating product, this solution acts as an operational efficiency multiplier. Automating user provisioning eliminates manual entry costs, and reducing the Average Handle Time (AHT) of incidents saves significant IT labor hours, translating to a strong positive Return on Investment (ROI).

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Import Setup	USN-1	As a System Admin, I want to configure a Data Source and generate an Import Set table (`import_employee`) so that the external CSV file can be staged.	2	High	SAIRAM M, KARPAGA VINAYAGAM S
Sprint-1	Data Transformation	USN-2	As a System Admin, I want to create a Transform Map targeting the User (`sys_user`) table so that staged data has a defined destination.	2	High	MANUNEETHI CHOLAN S
Sprint-1	Data Transformation	USN-3	As a System Admin, I want to use Auto Map and Mapping Assist to accurately link standard text fields and complex reference fields (Department, Manager) to the target table.	5	High	KARPAGA VINAYAGAM S
Sprint-2	Testing & Validation	USN-4	As a QA Tester, I want to run the Transform Map with a sample payload and verify that employee profiles are successfully inserted or updated without data loss.	3	High	SARAVANAN R
Sprint-2	Form Configuration	USN-5	As a System Admin, I want to modify the Incident form layout to include dot-walked fields (`Assigned to.department`, `Assigned to.manager.email`) so the data is visible on the UI.	3	High	RADAKRISHNAN S
Sprint-2	Incident Management	USN-6	As an IT Support Agent, I want the Incident form to automatically display the assignee's manager email and department when I assign a ticket, so I do not have to manually look up their profile.	1	Medium	SAIRAM M

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	9	16 Days	01 Mar 2026	15 Mar 2026		
Sprint-2	7	16 Days	16 Mar 2026	01 Apr 2026		

Velocity:

$$V = \frac{Sprint\ Duration}{Average\ Story\ Points} = \frac{16}{8} = 2$$

$$\therefore\ AV = \frac{Sprint\ Duration}{Velocity} = \frac{16}{2} = 8$$

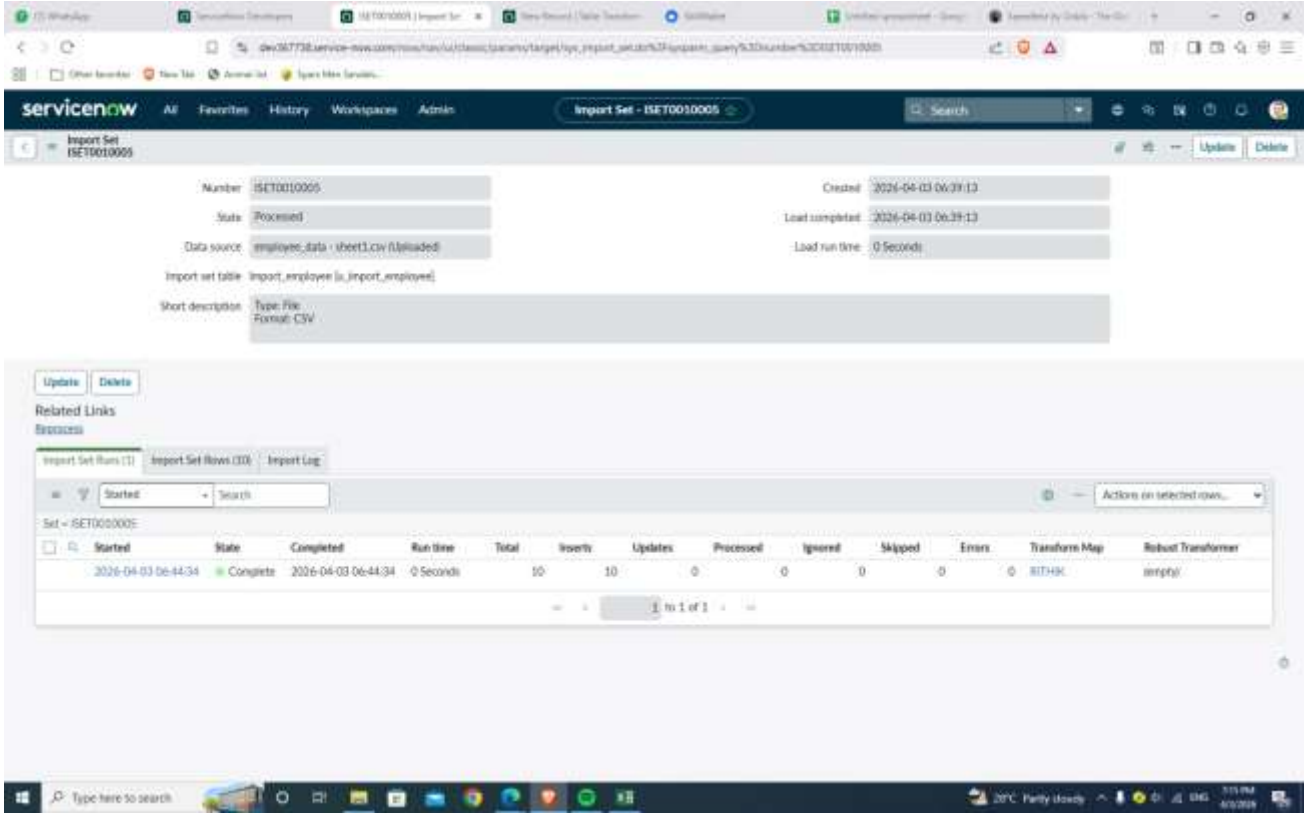
6. FUNCTIONAL AND PERFORMANCE TESTING

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Data Staging via Load Data (FR-1)	1. Navigate to Load Data. 2. Upload a valid CSV with 5 employee records. 3. Check `import_employee` table.	CSV is accepted; 5 rows are successfully created in the `import_employee` staging table.		
FT-02	Transform Map: String Fields (FR-2)	1. Run the Transform Map on the staged data. 2. Navigate to sys_user table. 3. Verify Name, Email, and Phone fields.	Name, Email, and Phone fields on the new `sys_user` records exactly match the CSV data.		
FT-03	Transform Map: Reference Fields (FR-2)	1. Open a newly imported user record in sys_user. 2. Check Department and Manager fields.	Department and Manager fields are successfully linked to the correct existing `sys_id` records, not just text strings.		
FT-04	Incident Form Auto-Population (FR-4)	1. Open a new Incident record. 2. Select one of the newly imported employees in the "Assigned to" field.	The employee's Department and Manager Email automatically appear on the form.		
RT-01	Error Handling for Missing References (NFR-3: Reliability)	1. Upload a CSV where one row has a Manager email that does not exist in the system. 2. Run Transform.	Transform completes. Valid rows are inserted. The invalid row logs an error (or ignores the manager field, based on config) but does not stop the batch.		
ST-01	Role-Based Access to Import (NFR-2: Security)	1. Impersonate a standard ITIL user. 2. Attempt to navigate to "Load Data" or "Transform Maps".	Access is denied. Module is hidden or restricted by ACLs. Only Admins/Import Admins have access.		

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
ST-02	UI Data Visibility (NFR-2: Security)	1. Impersonate an ITIL user. 2. View an Incident assigned to the new user.	ITIL user can view the dot-walked Department and Manager Email fields without triggering ACL read errors.		
UT-01	Dynamic Form Updates (NFR-1: Usability)	1. On an Incident, change the "Assigned to" user to person A, then to person B.	The dot-walked fields instantly update to reflect person B's details without requiring the user to save or reload the page.		
PT-01	Form UI Load Time (NFR-4: Performance)	1. Open an existing Incident with an assigned user. 2. Measure page load time.	The Incident form loads in under 2 seconds. The dot-walked fields do not cause noticeable UI lag.		
PT-02	Bulk Data Processing Limit (NFR-6: Scalability)	1. Upload a bulk CSV file containing 2,000 employee records. 2. Execute the Transform Map.	The platform processes all 2,000 records successfully within a reasonable timeframe (e.g., under 5 minutes) without timing out or degrading platform performance.		

7. RESULTS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Workspaces
Admin
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State
Processed
Data source
employee_data - sheet1.csv (Uploaded)
Import set table
Import_employee (u_import_employee)
Short description
Type: File
Format: CSV
Created
2026-04-03 06:39:13
Load completed
2026-04-03 06:39:13
Load run time
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Reprocess
Import Set Runs (1)
Import Set Rows (10)
Import Log
Started
Search
Actions on selected rows...
Set = ISET0010005
Started
State
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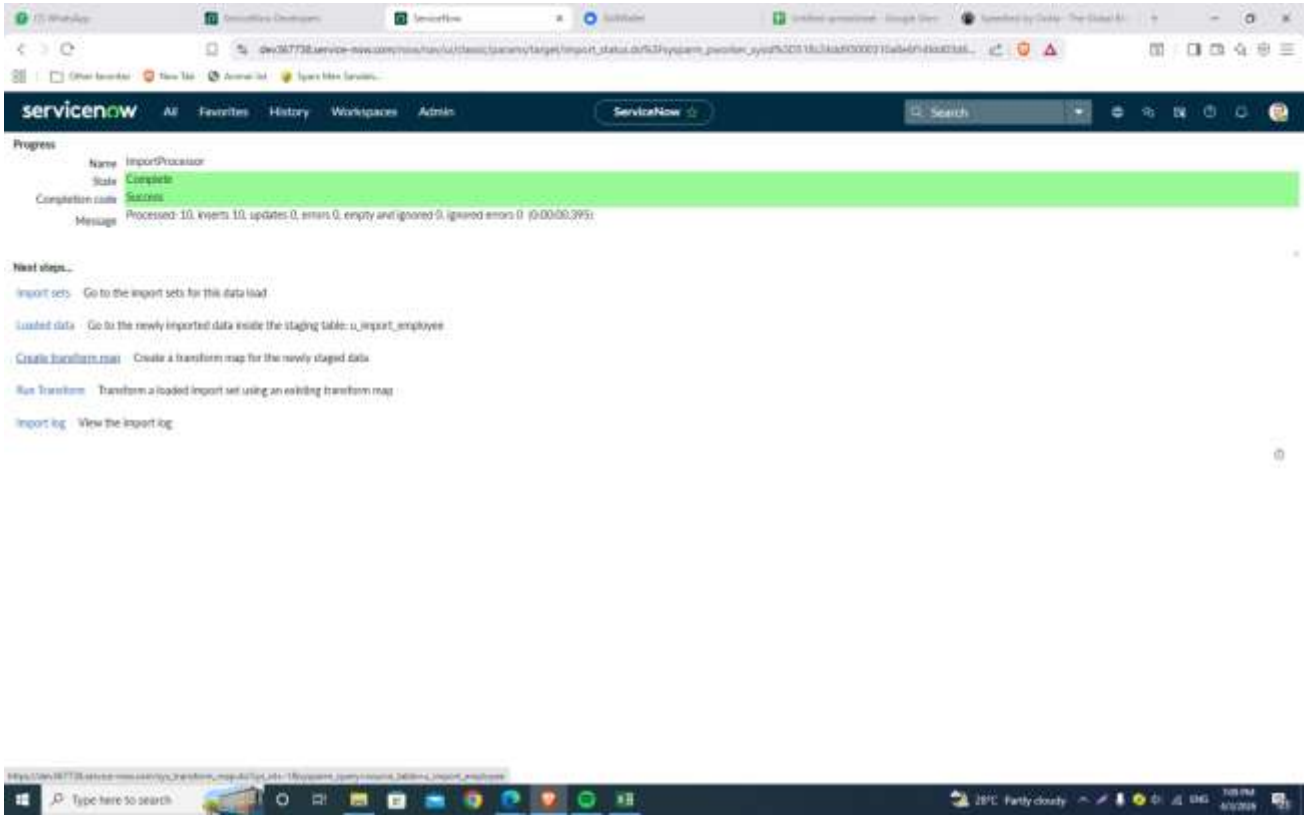

servicenow
All
Favorites
History
Workspaces
Admin
ServiceNow
Progress
Name
ImportProcessor
State
Complete
Completion code
Success
Message
Processed: 10, inserts: 10, updates: 0, errors: 0, empty and ignored: 0, ignored errors: 0 (0:00:00.395)
Next steps...
Import sets
Go to the import sets for this data load
Loaded data
Go to the newly imported data inside the staging table: u_import_employee
Create transform map
Create a transform map for the newly staged data
Run transform
Transform a loaded import set using an existing transform map
Import log
View the import log

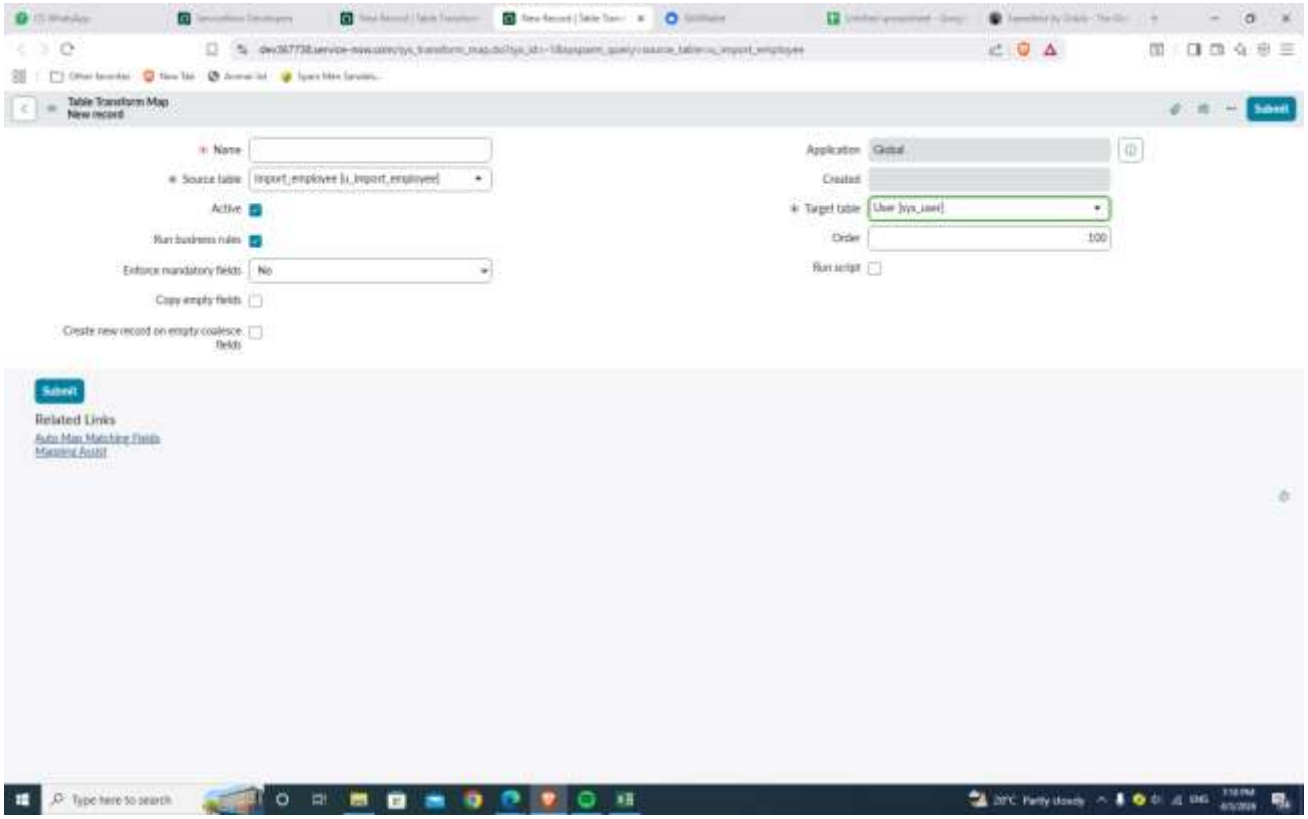
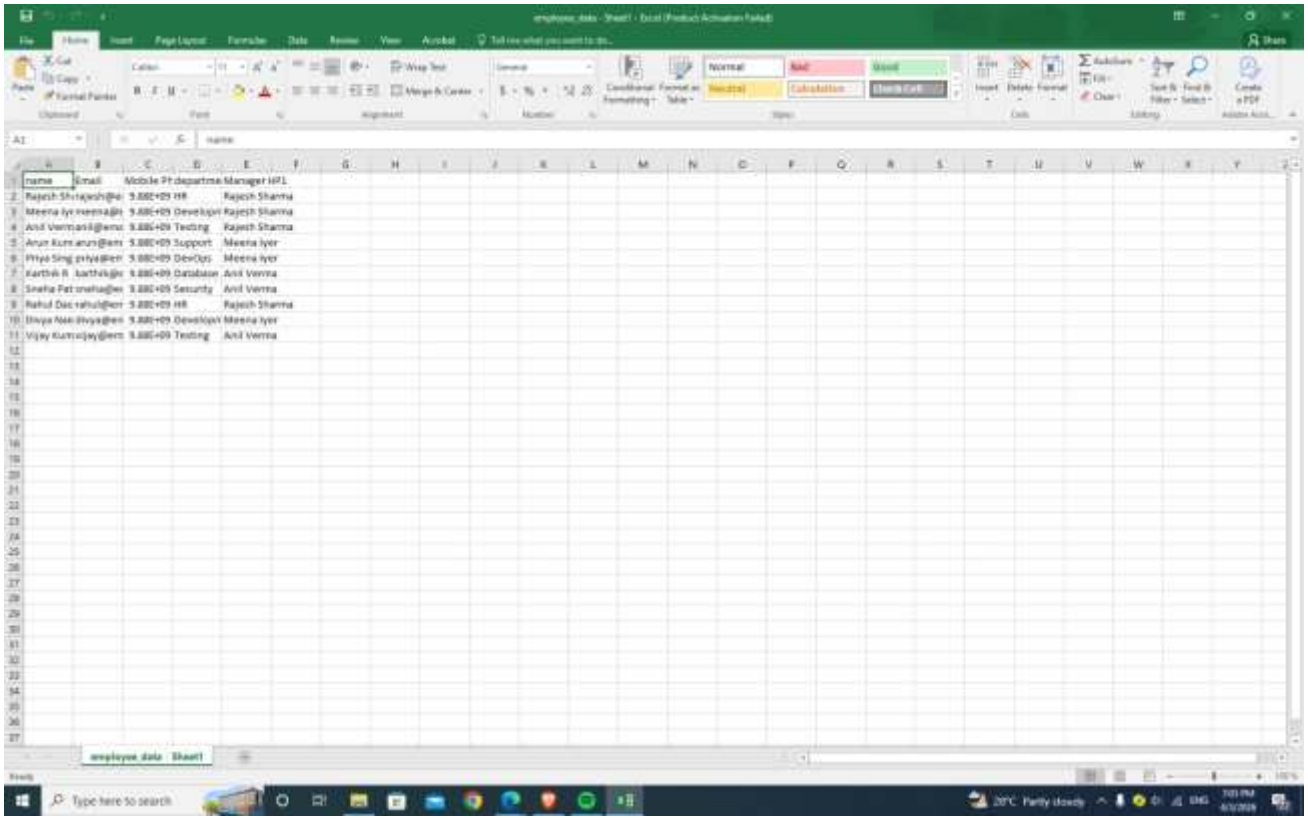


Table Transform Map
New record
Submit
Name
Source table
import_employee [u_import_employee]
Active
Run business rules
Enforce mandatory fields
No
Copy empty fields
Create new record on empty coalesce fields
Application
Global
Created
Target table
User [sys_user]
Order
100
Run script
Submit
Related Links
Auto Map Matching Fields
Mapping Assist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Manager HPL
Rajesh Shrivastava@ex 9.88E+09 HR Rajesh Sharma
Meena Iyer meena@ex 9.88E+09 Developm Rajesh Sharma
Anil Verma anil@exa 9.88E+09 Testing Rajesh Sharma
Arun Kumar arun@exa 9.88E+09 Support Meena Iyer
Priya Singh priya@ex 9.88E+09 DevOps Meena Iyer
Karthik R karthik@ex 9.88E+09 Database Anil Verma
Sneha Patel sneha@ex 9.88E+09 Security Anil Verma
Rahul Das rahul@ex 9.88E+09 HR Rajesh Sharma
Divya Nair divya@ex 9.88E+09 Developm Meena Iyer
Vijay Kumar vijay@ex 9.88E+09 Testing Anil Verma
employee_data - Sheet1

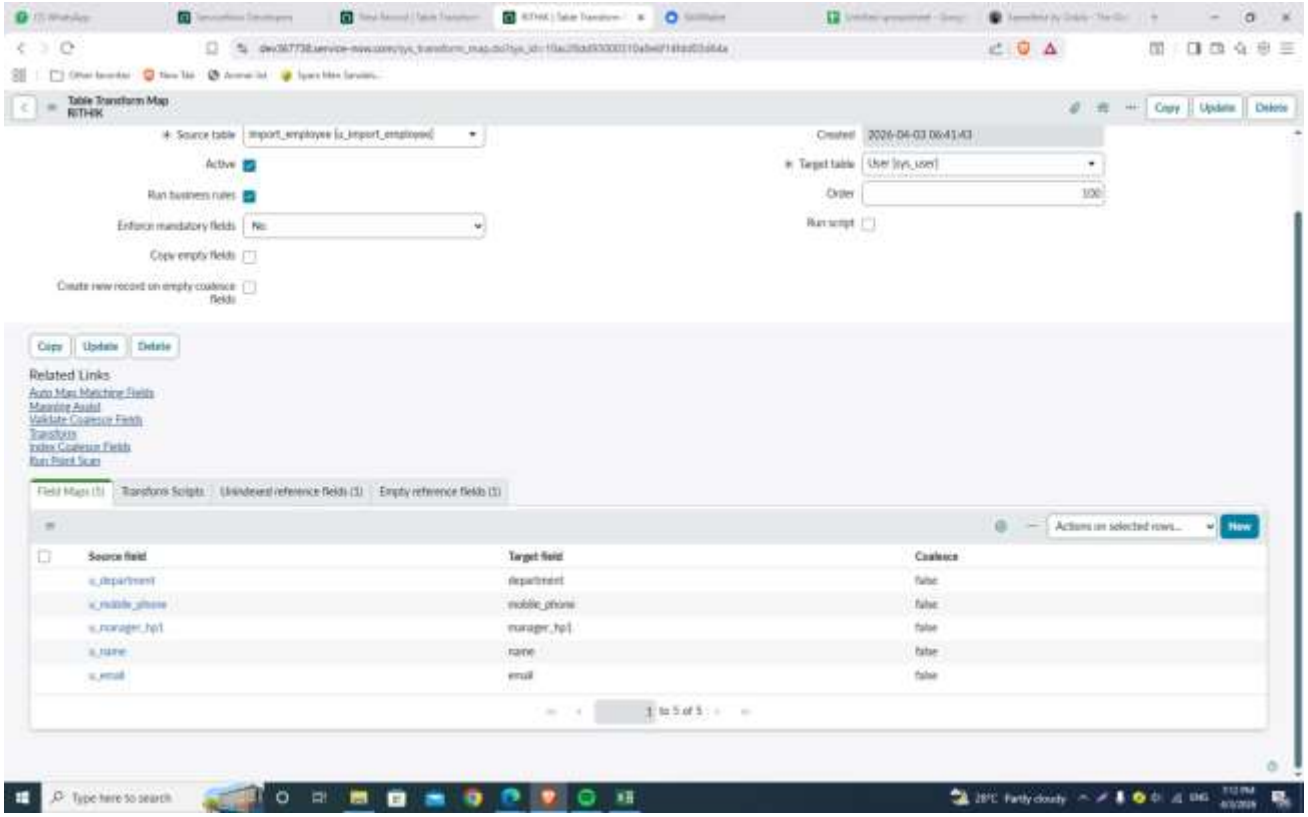
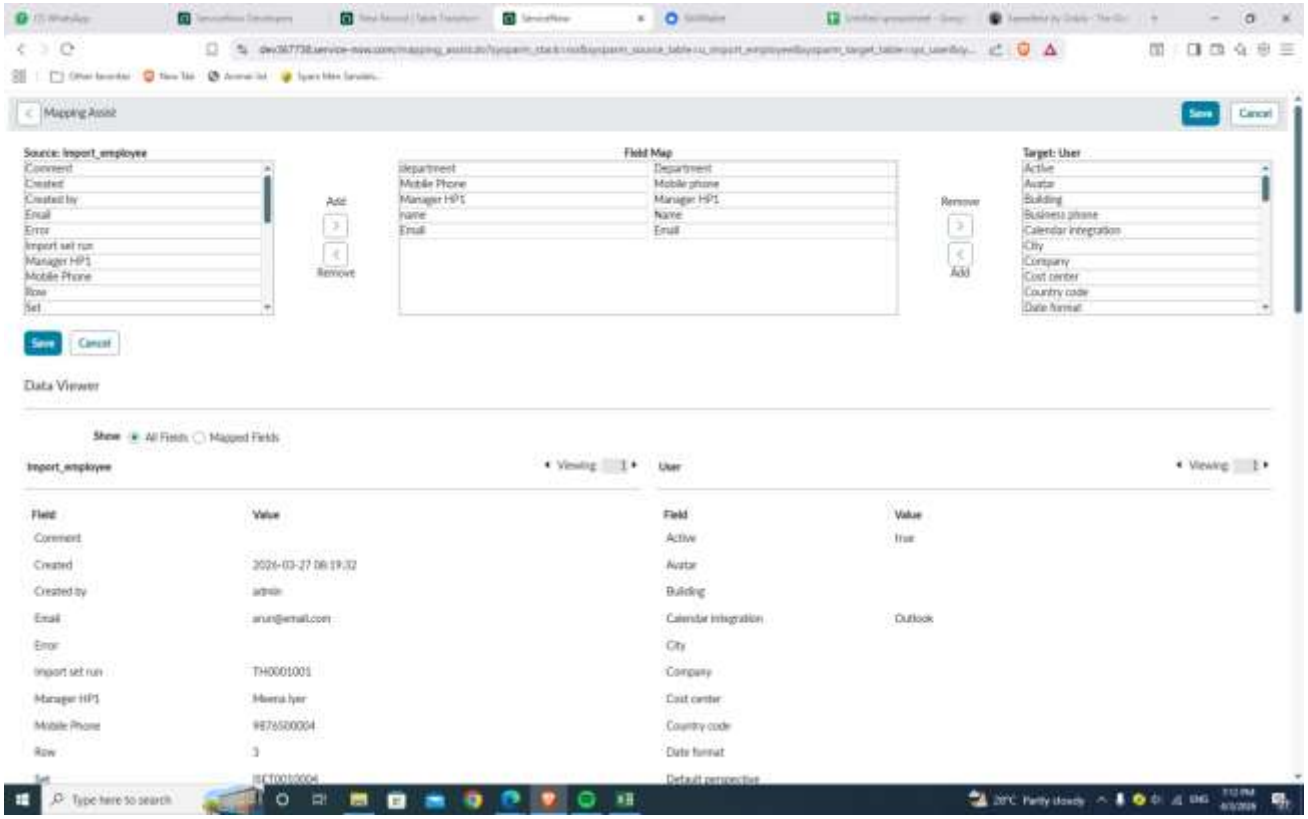
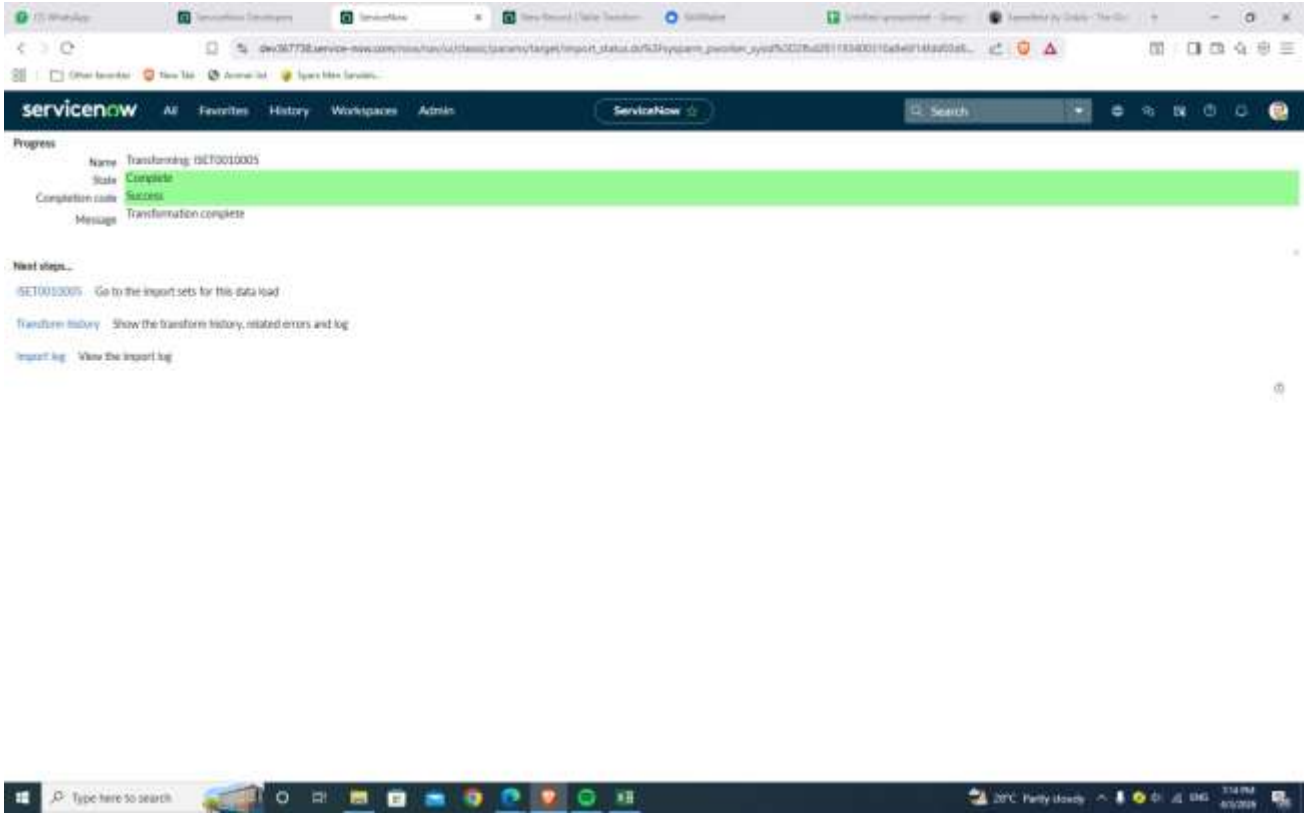


Table Transform Map
RITHIK
Copy
Update
Delete
Source table
import_employee [u_import_employee]
Created
2026-04-03 06:41:43
Active
Target table
User [sys_user]
Run business rules
Order
100
Enforce mandatory fields
No
Run script
Copy empty fields
Create new record on empty coalesce fields
Related Links
Auto Map Matching Fields
Mapping Assist
Validate Coalesce Fields
Transform
Index Coalesce Fields
Run Point Scan
Field Maps (5)
Transform Scripts
Unindexed reference fields (1)
Empty reference fields (1)
Actions on selected rows...
New
Source field
Target field
Coalesce
u_department
department
false
u_mobile_phone
mobile_phone
false
u_manager_hp1
manager_hp1
false
u_name
name
false
u_email
email
false
1 to 5 of 5


Mapping Assist
Save
Cancel
Source: import_employee
Comment
Created
Created by
Email
Error
Import set run
Manager HP1
Mobile Phone
Row
Set
Add
Remove
Field Map
department
Department
Mobile Phone
Mobile phone
Manager HP1
Manager HP1
name
Name
Email
Email
Remove
Add
Target: User
Active
Avatar
Building
Business phone
Calendar integration
City
Company
Cost center
Country code
Date format
Save
Cancel
Data Viewer
Show
All Fields
Mapped Fields
import_employee
Viewing 1
User
Viewing 1
Field
Value
Comment
Created
2026-03-27 06:19:32
Created by
admin
Email
arun@email.com
Error
Import set run
TH0001001
Manager HP1
Meena Iyer
Mobile Phone
9876500004
Row
3
Set
IECT0010004
Field
Value
Active
true
Avatar
Building
Calendar integration
Outlook
City
Company
Cost center
Country code
Date format
Default perspective


servicenow
All
Favorites
History
Workspaces
Admin
ServiceNow
Progress
Name Transforming: ISET0010005
State Complete
Completion code Success
Message Transformation complete
Next steps...
ISET0010005 Go to the import sets for this data load
Transform history Show the transform history, related errors and log
Import log View the import log

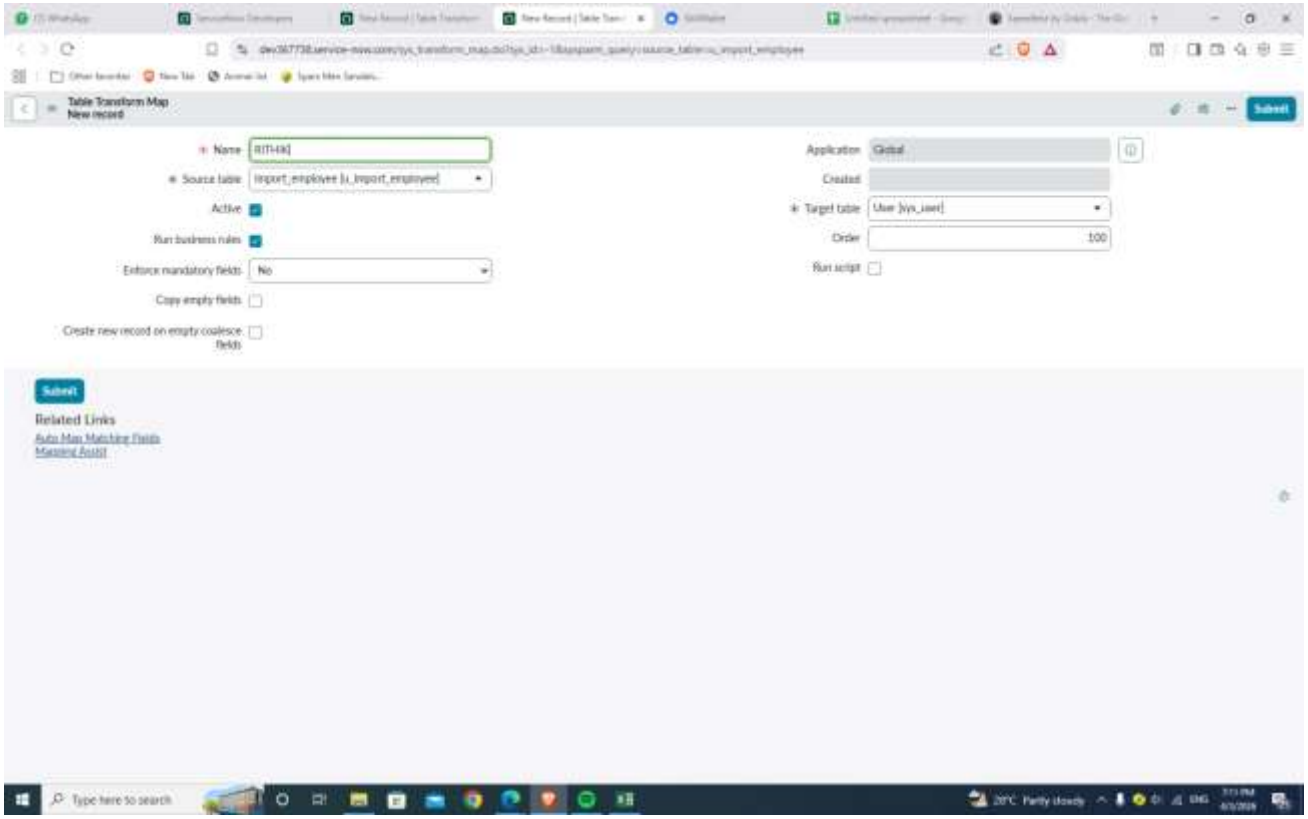
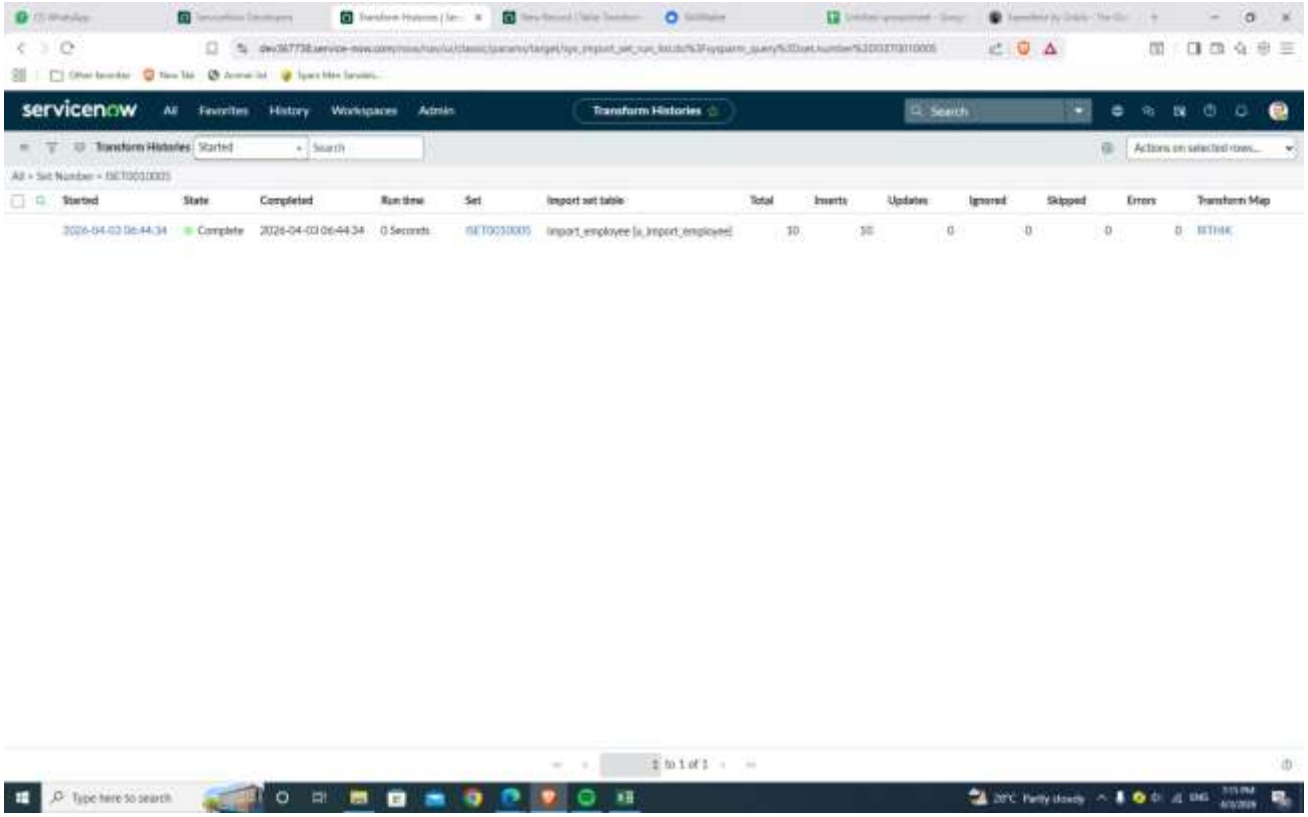
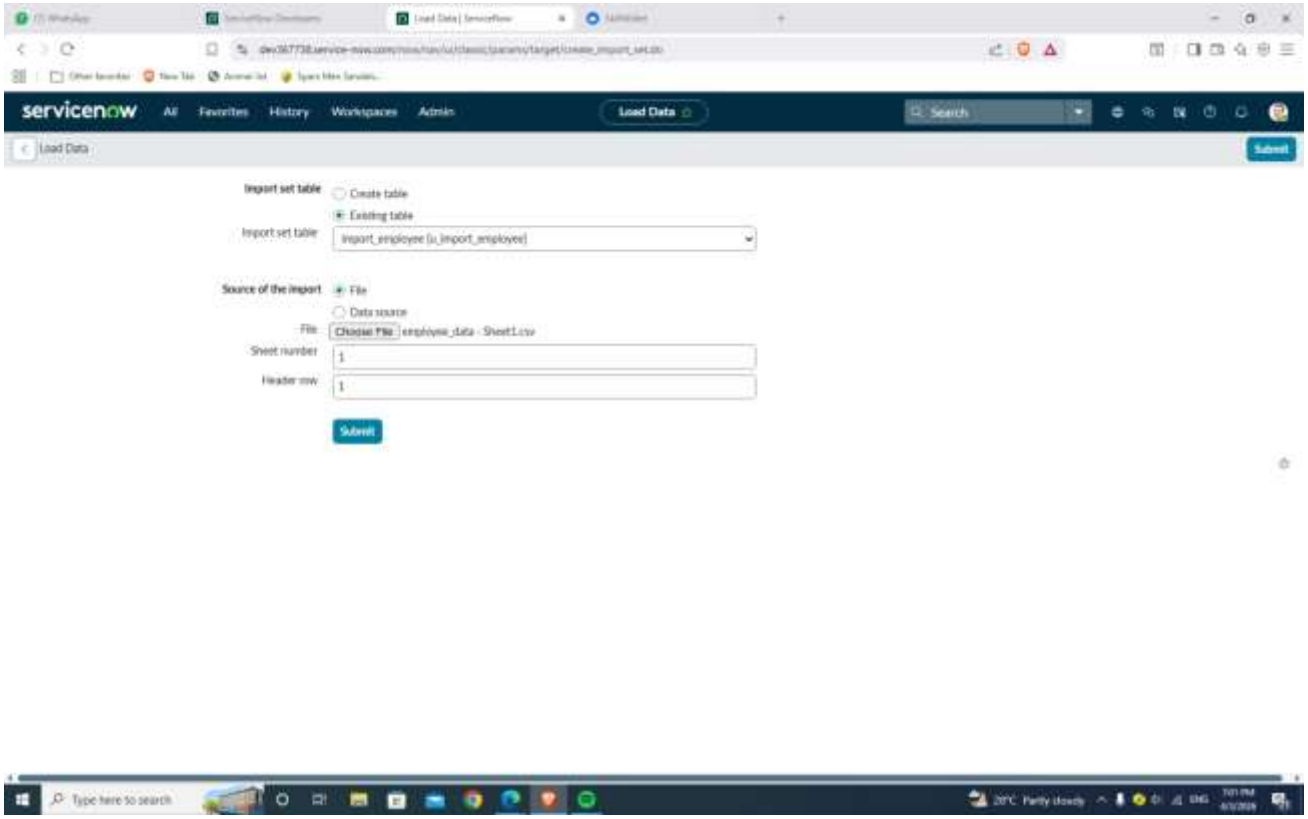
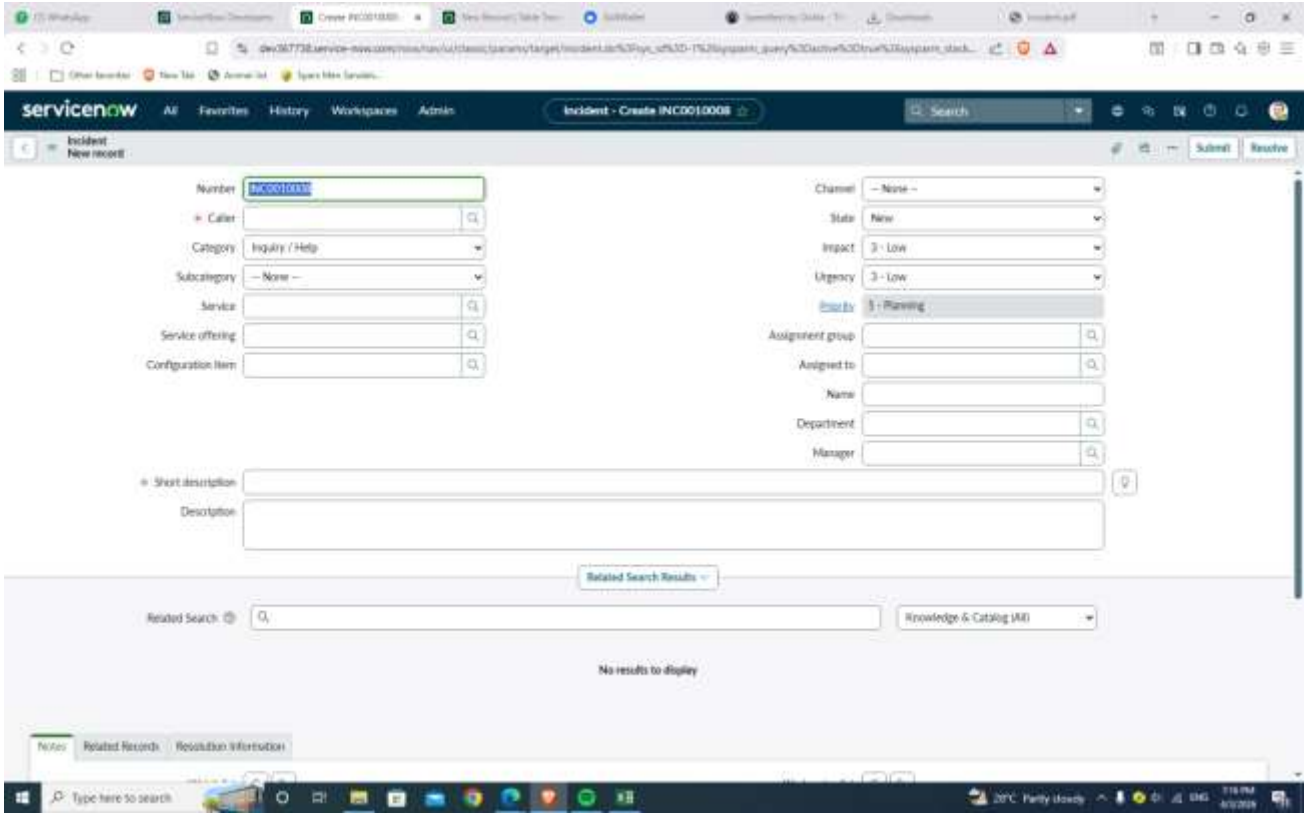
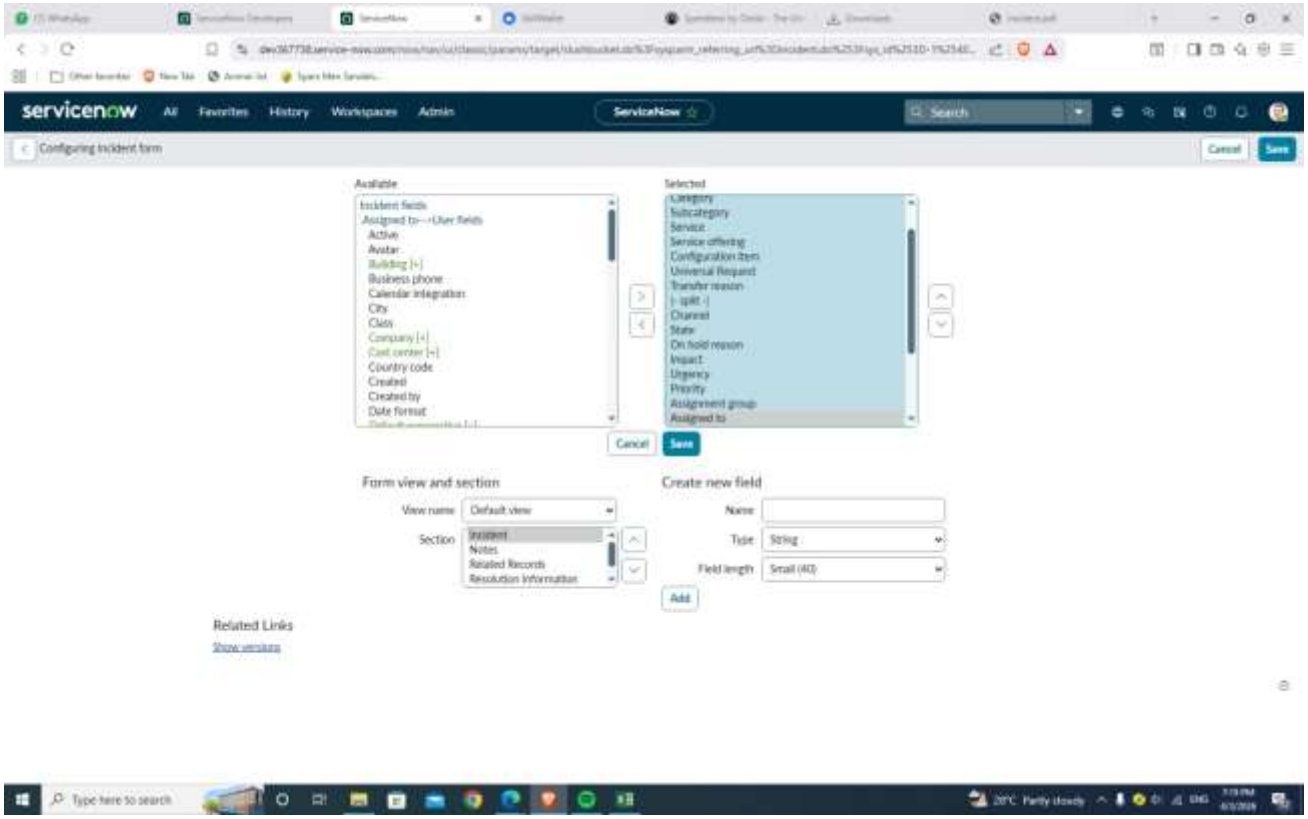


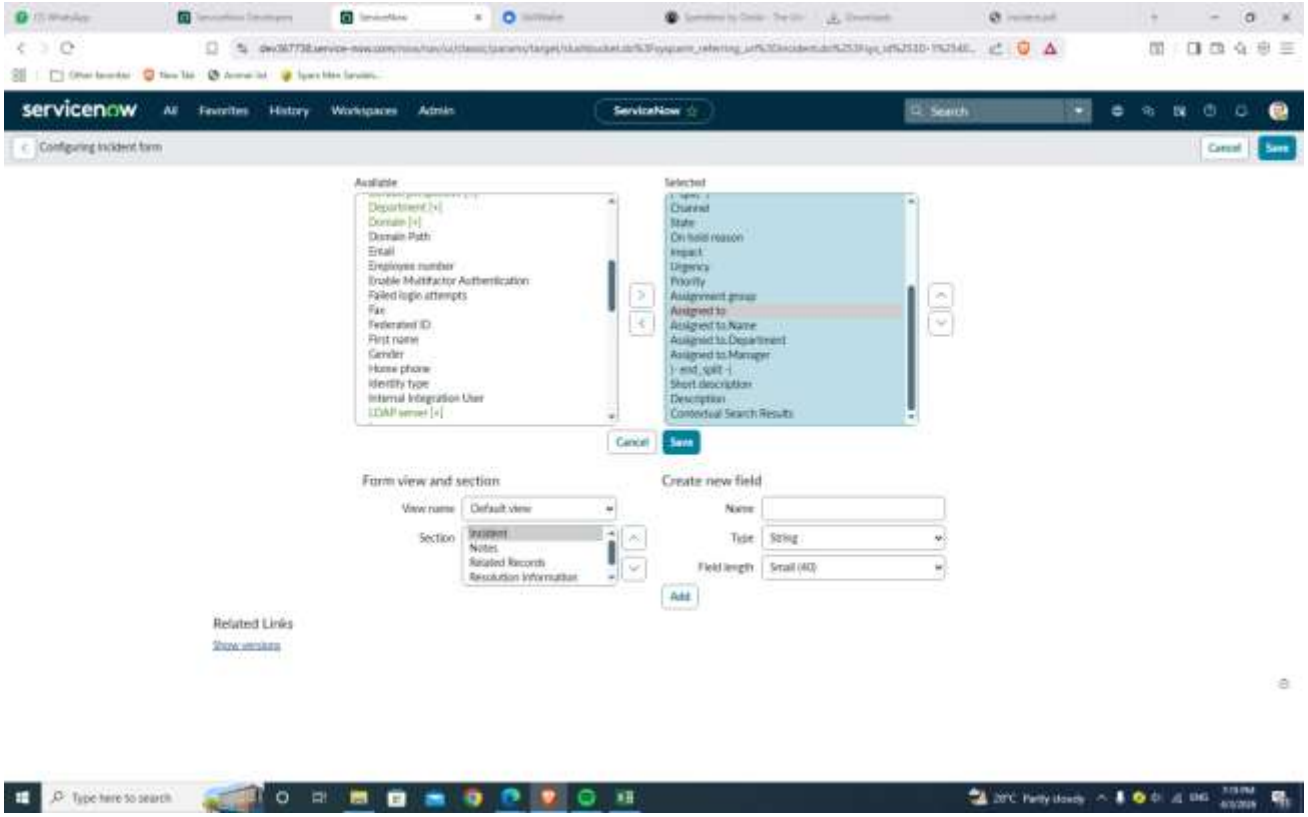
Table Transform Map
New record
Submit
Name RITHIK
Source table import_employee [u_import_employee]
Active
Run business rules
Enforce mandatory fields No
Copy empty fields
Create new record on empty coalesce fields
Application Global
Created
Target table User [sys_user]
Order 100
Run script
Submit
Related Links
Auto Map Matching Fields
Mapping Assist


servicenow
All
Favorites
History
Workspaces
Admin
Transform Histories
Transform Histories Started
Actions on selected rows...
All > Set Number = ISET0010005
Started
State
Completed
Run time
Set
Import set table
Total
Inserts
Updates
Ignored
Skipped
Errors
Transform Map
2026-04-03 06:44:34
Complete
2026-04-03 06:44:34
0 Seconds
ISET0010005
import_employee [u_import_employee]
10
10
0
0
0
0
IETHIK
1 to 1 of 1
Type here to search


servicenow
All
Favorites
History
Workspaces
Admin
Load Data
Load Data
Submit
Import set table
Create table
Existing table
Import set table
import_employee [u_import_employee]
Source of the import
File
Data source
File
Choose File
employee_data - Sheet1.csv
Sheet number
1
Header row
1
Submit
Type here to search


servicenow
All
Favorites
History
Workspaces
Admin
Incident - Create INC0010008
Incident
New record
Submit
Resolve
Number
INC0010008
Caller
Category
Inquiry / Help
Subcategory
-- None --
Service
Service offering
Configuration item
Channel
-- None --
State
New
Impact
3 - Low
Urgency
3 - Low
Priority
5 - Planning
Assignment group
Assigned to
Name
Department
Manager
Short description
Description
Related Search Results
Related Search
Knowledge & Catalog (All)
No results to display
Notes
Related Records
Resolution Information


servicenow
All
Favorites
History
Workspaces
Admin
ServiceNow
Configuring incident form
Cancel
Save
Available
Selected
Incident fields
Assigned to--User fields
Active
Avatar
Building [+]
Business phone
Calendar integration
City
Class
Company [+]
Cost center [+]
Country code
Created
Created by
Date format
Category
Subcategory
Service
Service offering
Configuration item
Universal Request
Transfer reason
[- split -]
Channel
State
On hold reason
Impact
Urgency
Priority
Assignment group
Assigned to
Cancel
Save
Form view and section
View name
Default view
Section
Incident
Notes
Related Records
Resolution Information
Create new field
Name
Type
String
Field length
Small (40)
Add
Related Links
Show versions


servicenow
All
Favorites
History
Workspaces
Admin
ServiceNow
Search
Configuring incident form
Cancel
Save
Available
Department [+]
Domain [+]
Domain Path
Email
Employee number
Enable Multifactor Authentication
Failed login attempts
Fax
Federated ID
First name
Gender
Home phone
Identity type
Internal Integration User
LDAP server [+]
Selected
Channel
State
On hold reason
Impact
Urgency
Priority
Assignment group
Assigned to
Assigned to Name
Assigned to Department
Assigned to Manager
) end_split -)
Short description
Description
Contextual Search Results
Cancel
Save
Form view and section
View name
Default view
Section
Incident
Notes
Related Records
Resolution Information
Create new field
Name
Type
String
Field length
Small (40)
Add
Related Links
Show versions

8. ADVANTAGES & DISADVANTAGES

8.1. Advantages

1. Significant Time Savings (Reduced AHT): By using dot-walking on the Incident form, IT Support Agents no longer need to navigate away from their active screen to search for a user's department or manager. This immediately reduces the Average Handle Time (AHT) for ticket routing and escalation.
2. Elimination of Manual Effort: Automating the data ingestion via Import Sets replaces the tedious and error-prone process of manually typing in new employee profiles one by one.
3. Highly Maintainable (Low-Code/No-Code): The entire solution relies on native platform capabilities. Because it uses standard Transform Maps and Form Design (dot-walking) instead of complex custom GlideRecord scripts or GlideAjax calls, the system is much easier to maintain.
4. Upgrade Safe: Native platform configurations are thoroughly tested by ServiceNow. When your organization upgrades to a new ServiceNow release family, this automated pipeline and form layout will carry over safely without breaking.
5. Scalability: The Import Set staging architecture is specifically designed to handle large volumes of data. Whether you are importing 10 employees a week or 5,000 employees during a corporate merger, the Transform Map engine will process them efficiently in the background.
6.

8.2. Disadvantages & Limitations

1. Total Reliance on Source Data Quality: The automation is only as good as the CSV file provided by the external HR system. If the HR system provides an incorrect manager email, or slightly misspells a department name, the Transform Map will either fail to link the reference or create duplicate/incorrect data in ServiceNow.
2. Not Strictly Real-Time: Unless the CSV upload is triggered by an active REST API call, Import Sets are usually batch-processed (e.g., uploaded manually or scheduled to run via FTP once a day). This means there could be a slight delay between an employee being added to HR and appearing in ServiceNow.
3. Potential UI Clutter: While dot-walking is powerful, adding too many dot-walked fields (e.g., Manager's Phone, Department Head, Cost Center) can quickly clutter the Incident form and overwhelm the agent with unnecessary information.
4. Read-Only Data on the Target Form: Dot-walked fields displayed on the Incident form are read-only. If a support agent notices that an employee's department is incorrect while working on an incident, they cannot fix it directly on the Incident form; they must navigate to the actual `sys_user` record (or fix it in the source HR system) to update it.
5. Initial Configuration Complexity: While it requires no code, setting up Mapping Assist correctly for complex reference fields (like ensuring it matches the Manager by *email* and not just by *first name*) requires a solid understanding of relational database structures and ServiceNow administration.

9. CONCLUSION

The **"Automating Data Import and Relationship Mapping"** project has successfully modernized both the data management and IT support workflows within our ServiceNow environment. By transitioning from manual data entry to an automated pipeline using **System Import Sets** and **Transform Maps**, the organization has established a highly scalable, error-free method for ingesting external employee data and accurately mapping complex reference relationships.

Furthermore, leveraging native **Dot-Walking** capabilities on the Incident form has delivered immediate, tangible benefits to the IT Support team. Support Agents now have instant, zero-click visibility into crucial contextual information—specifically the assigned employee's Department and Manager—directly on the active ticket. This enhancement eliminates unnecessary navigation, directly contributing to a lower **Average Handle Time (AHT)** and a more streamlined incident escalation process.

Ultimately, this project demonstrates the strategic value of maximizing ServiceNow's out-of-the-box, low-code features. Because the solution avoids complex custom scripting, the resulting architecture is highly maintainable, secure, and upgrade-safe. The successful deployment of this automation not only solves the immediate operational bottlenecks but also sets a strong, resilient foundation for future automation initiatives across the platform.

10. FUTURE SCOPE

While the current implementation successfully automates manual data entry and improves UI visibility via Import Sets and Dot-Walking, the project establishes a foundation that can be expanded in several strategic directions:

1. Direct API Integration (Eliminating CSVs)

The most immediate future enhancement is to move away from batch CSV uploads. By leveraging ServiceNow Integration Hub or custom REST APIs, the platform can be connected directly to the external HR Information System (e.g., Workday, SAP SuccessFactors). This will enable true real-time, bi-directional synchronization of employee data the moment a change is made in HR.

2. Cross-Module UI Expansion

Currently, the dot-walking configuration is limited to the Incident form. This contextual visibility can be expanded across all core IT Service Management (ITSM) modules. Future phases should include adding Department and Manager fields to Requested Items (RITMs), Problem records, and Change Requests, ensuring uniform context for all IT workflows.

3. Automated Provisioning Workflows

With accurate, automated relationship mapping now reliably establishing an employee's Department and Manager in the `sys_user` table, this data can be used as a trigger for automation. Future scope includes building workflows that automatically generate catalog tasks for hardware, software, and system access provisioning tailored precisely to the new user's department the moment their profile is imported.

4. Intelligent Escalations & Notifications

The newly established Manager relationships can be utilized to build smarter notification triggers. For example, the system can be configured to automatically CC an employee's manager when a Priority 1 (Critical) incident is opened for them, or build automated approval routing for service requests based directly on the mapped manager.

5. Advanced Data Validation & Quarantine

To further protect data integrity, future iterations can implement robust Pre-Transform Scripts. These scripts would validate data quality (e.g., checking for correct email syntax or flagging if a department doesn't exist) *before* the transform runs. Erroneous records would be automatically diverted into a "quarantine" queue for an administrator to review, while clean records process normally.

11. APPENDIX

11.1 Sample Data Set Link

The sample dataset consists of multi-level hierarchical employee data with managers.
Link: **https://github.com/Vinothedeveloper/Automating-Data-Import-and-Relationship-Mapping-using-Import-Sets-and-Dot-Walking/blob/main/employee_data - Sheet1.csv**

11.2 Project Demonstration Link

We have attached the Project demonstration link down below. It explains the project's architecture and the key design decisions made during the project planning and execution.

Link: **https://drive.google.com/file/d/1Yz1KNCs5wQzo8DGrA4bCfjAsFjvP1ug57/view**

11.3 GitHub Link

The project details and the phases of development of the project have been uploaded to GitHub. We have attached the link to our GitHub repo below.

Link: **https://github.com/Vinothedeveloper/Automating-Data-Import-and-Relationship-Mapping-using-Import-Sets-and-Dot-Walking.git**